Somatosensory system

(~ TACTO)

Somatic
- Proprioception – skeletal muscle, skin, joint
- Exteroception – contact, pressure, motion, vibration, temperature, pain. (nociception)
- Interoception – not conscious, mostly chemical

Primary Sensory Neurons Clustered in Dorsal Root Ganglia

2 functions
- transduction & encoding into electrical signals
- transmission to SNC

Soma → Ganglio Dorsal ⇒ Pseudo-unipolar $\xrightarrow{\text{adult}}$ Bipolar
- 1 Pole ⇒ SNC → 1st Synapsis (Primary afferent fiber)
- 1 Pole ⇒ Periphery

Peripheral Somatosensory Nerve Fibers

	Fiber diameter (μm)	Conduction v (m/s)	Muscle nerve	Cutaneous Nerve
Myelinated				
large diam.	12-20	72-120	I Skeletal muscle	Aα
medium	6-12	36-72	II	Aβ
small	1-6	4-36	III	Aδ
Unmyelinated	0,2-1,5	0,4-2.0	IV	C

each ≠ place

burning pain

Cuando se estimula un axón pequeño (los que tienen ↑ Resistencia) se percibe como doloroso.

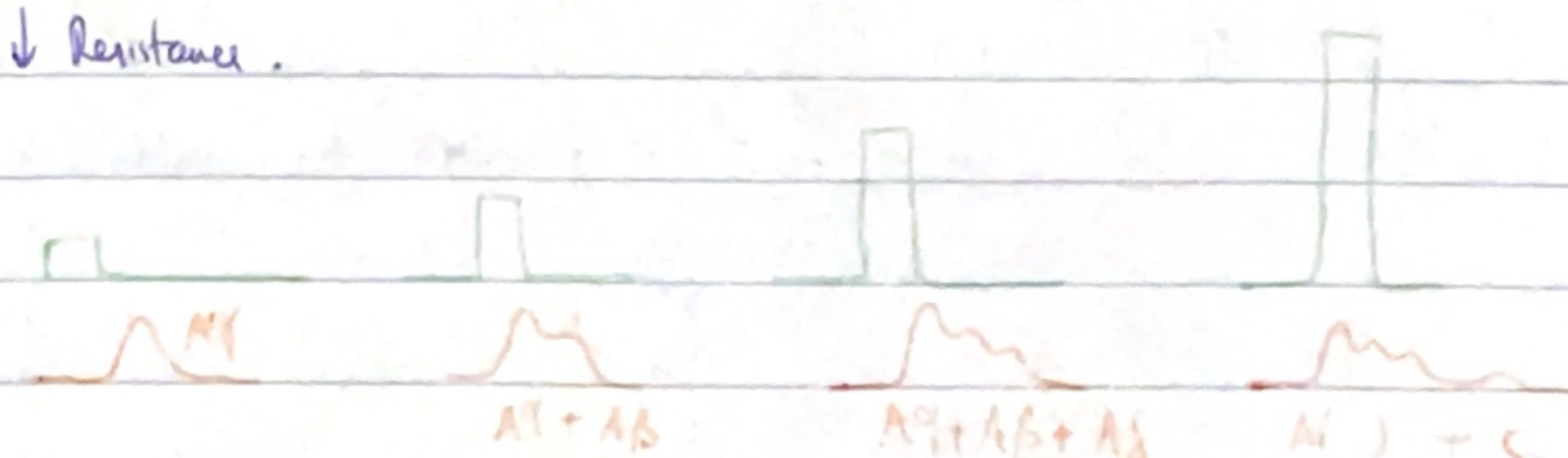
como van sin lentas pueden sumar estímulo continuo

Σ nerve fibers = compound action potential

bigger the axon ⇒ ↓ Resistance.

Stimulus Intensity

Nerve response



Sensation evoked	Barely perceptible	tingling	pain	burning

Specialized Receptors

(8 types) → Mechanoreceptors mediate touch & proprioception

They open the ion channels stretch-sensitive allowing Na^+ and Ca^{++} to depolarize

mechanical → Distention & Muscle Stretching ⇉ Physical deformation

Some use ⎧ direct activation on the ion channel
⎩ 2nd messenger $\xrightarrow{\text{tells}}$ the ion channel Slower but amplifies it (PAIN)

TÓNICOS

Slowly Adapting (SA): continue to fire in response to steady pressure on the skin

- Merkel (SA1) – $\vec{P}$ → edges, corners, points, texture (Braille)
- Ruffini (SA2) – Stretching ⎧ Shape of large objects
  ⎩ movements (fingers, joints)

FÁSICOS

Rapidly Adapting (RA) – Motion (Also ≠ from SA in location and size)

Not in hairy skin • Meissner (RA1) – low-frequency – initial contact

- Pacini (RA2) → high-frequency vibration (mostly in objects)

hairy skin receptors.

Mechanoreceptors that measure muscle and joint positions.

→ Muscle spindle – Intrafusal fibers ⎧ detect stretch
⎩ also innervated by motor neurons

Principal mediating PROPIOCEPTION

→ Golgi tendon – forces generated by muscle contraction UNCONSCIOUS

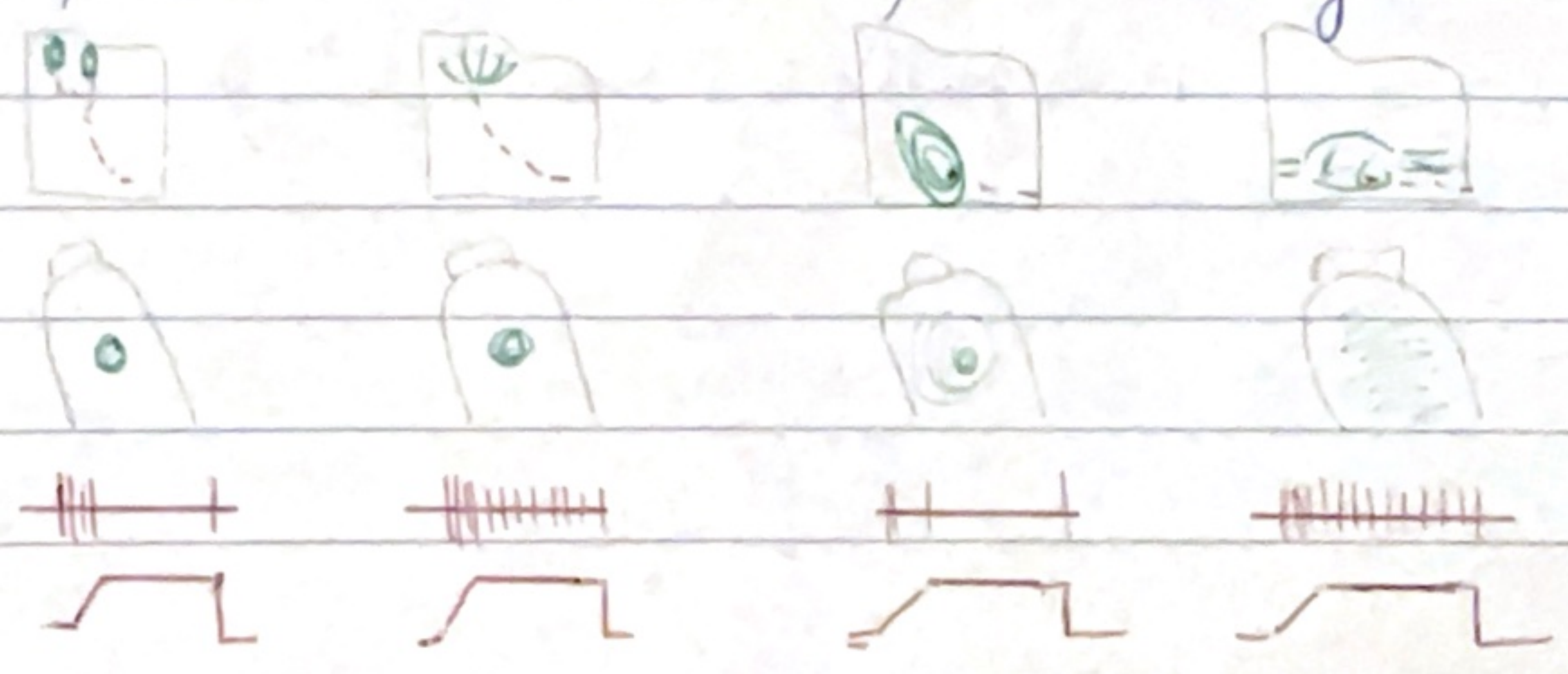
	Meissner corpuscle	Merkel cells	Pacinian corpuscle	Ruffini endings

Receptors

Receptive field

Neural spike train

stimulus



Nociceptors

Mechanical & thermal -

$A\delta \rightarrow$ short latency, sharp and pricking (Mechanical mostly) — pinpoint

$C \rightarrow$ dull, burning pain, diffuse, poorly localized $\rightarrow$ polymodal $\rightarrow$ mechanical, thermal, chemical

Thermal

humans : cold, cool, warm, hot

31 – 36° $\rightarrow$ we don't notice ≠

<31° $\rightarrow$ Cool   <10-15 Pain

>36 $\rightarrow$ Warm   >45° "

6 ≠ fibers:

low-threshold } cold receptor

high " } warm "

+

2 classes of heat receptors.

Warm receptors are like linear thermometers
↓
humans are more sensitive to cold that warmness

low + cold $\rightarrow$ sudden drops

high + cold $\rightarrow$ signal rapid skin cooling

Itch

Recent discoveries $\rightarrow$ type C fiber (low $\vec{v}$)

⇒ Skin and deeper tissues innervated by afferent fibers of a single spinal nerve $\rightarrow$ DERMATOME

Muscles $\rightarrow$ MIOTOME

(one)

- 4/12 Cranial nerves are MIXED
- 30/31 Spinal " " " (exception = C1)

Lateral	Medial
A_δ	$A\alpha$
C	$A\beta$
	↓
	dorsal columns

Spinal Gray matter $\rightarrow$ 10 layers $\rightarrow$ I to X (dorsal $\rightarrow$ ventral)

The somatotopic map is inverted in the medial lemniscus.

Spinothalamic System tract → principal pathway transmitting noxious, thermal and visceral information

formed of laminae I, V, VII. (formal, pain; >50%)

Most (82%) cross the midline and ascend contralateral

12% ipsilateral

6% bilateral

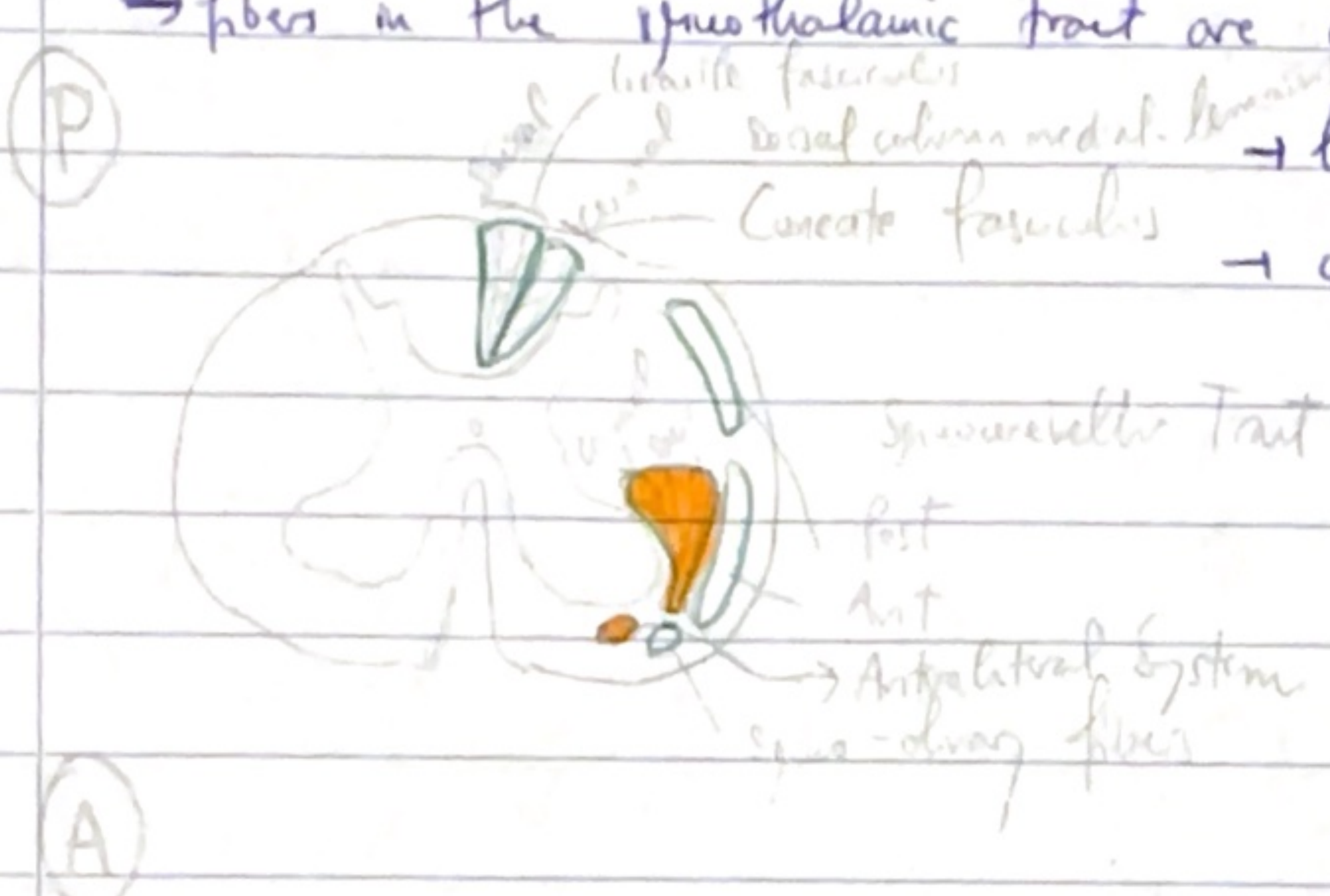
Noxious and thermal information → contralateral

Touch and proprioception → ipsilateral

⇒ fibers in the spinothalamic tract are organised somatotopically

→ lumbar and sacral → laterally

→ cervical → medially



Lamina I cross midline (temp)

(contra) lateral spinothalamic tract

lamina V (pain intensity)

(contra) ventral " "

Lamina VII

lam. VII (emotional response to stimuli)

2 vías: ant → propiocepción

lat → pain & temp

⇒ 3 neuronas desde piel hasta corteza.

1) piel - médula (seudounipolar)

2) sinapsis con neurona 2ria en sust Gelatinosa Rolando.

núcleo propio

|

cruzan al otro lado (comisura ant)

→ [illegible]

3) sinapsis con neurona del tálamo

Thalamus

↓

Specialized Somatosensory Regions

Small diameter axons reach the cerebral cortex through other pathways: spinomesencephalic, spinoreticular, spinobulbar..

↳ mostly to hypothalamus & amygdala

→ homeostasis and autonomic function